MODULE 1

1. a) Identify the different types of system calls and relate each type to appropriate operating-system operations.

   b) Describe the different types of computer-system architectures used in modern operating system.

2. a) Identify the process concept and explain how process states and process operations are related during process execution.

   b) Interview the abstract view of a computer system with a neat diagram and show how its major components interact with the operating system.

MODULE 2

3. a) Consider the following set of process, with the length of CPU burst given in milliseconds

Process	Burst time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

   The process are assumed to have arrived in order P1, P2, P3, P4 and P5 all at time 0

   a) Draw Gantt Chart that illustrates execution of this process using the following scheduling algorithm: FCFS, NON-Preemptive SJF, non-preemptive priority and Round Robin (Quantum=1).
   b) What is the turnaround time of each process of the scheduling algorithms in Part a?
   c) What is the waiting timer of each process for each of this scheduling algorithms? Which of the algorithms results in the minimum average waiting time (over all process)?

   b) Demonstrate the handling of threading issues such as fork(), exec(), signal handling, thread cancellation, and thread-local storage using suitable examples.

4. a)Compare the many-to-one, one-to-one, and many-to-many multithreading models with respect to their execution characteristics.

   b) Consider the following snapshot of a system

Process	**Arrival Time**	**Burst time**
P0	0	3
P1	2	6
P2	4	4
P3	6	5
P4	8	2

Consider the following set of processes with CPU burst time(in ms). Compute the waiting time and average turnaround time forthe above process using FCFS, SJF non preemptive scheduling algorithm.

MODULE 3

5. a) Illustrate a classic software-based solution to the critical-section problem.
   b) Analyze the Readers-Writers problem and Dining-Philosophers problem with respect to process synchronization.